POLYNOMIAL IDENTITIES AND INTEGRALITY FOR INVARIANTS OF A FINITE GROUP ACTION MODULO AN IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Invariant elements b of (R/JR)^G lift to elements of R^G modulo the ideal generated by J, with a monic polynomial relation. invoked at proof line 32619. Supplies part (1).
- For x in R^G the polynomial prod (T - sigma(x)) lies in the ideal generated by T - x and hence its reduction modulo J is determined by x mod J. invoked at proof lines 32620-32622. Gives (T - a)^n = T^n in part (2).
- An integral ring map inducing purely inseparable residue field extensions induces homeomorphisms on spectra. invoked at proof lines 32624-32625. Gives the final spectral statement.

Source reference. Original lemma 0BRF: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32540–32540. Cited at proof line 32619 with I = JR for part (1); the conclusion is used inside the counted span.

Source reference. Original lemma 0H34: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32569–32569. Cited at proof lines 32622-32623 for the congruence (T - x)^n = T^n modulo J; the conclusion is used inside the counted span.

Source reference. Original lemma 09EF: sources/stacks-chow/tags.txt, source lines 11898–11898. Cited at proof lines 32624-32625 for the spectral homeomorphisms; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let G be a finite group of order n acting on R. Let $J \subset R^G$ be an ideal. Then $R^G/J \to (R/JR)^G$ is ring map such that*

(1) *for $b \in (R/JR)^G$ there is a monic polynomial $P \in R^G/J[T]$ whose image in $(R/JR)^G[T]$ is $(T-b)^n$,*
(2) *for $a \in \mathrm{Ker}(R^G/J \to (R/JR)^G)$ we have $(T-a)^n = T^n$ in $R^G/J[T]$.*

In particular, $R^G/J \to (R/JR)^G$ is an integral ring map which induces homeomorphisms on spectra and purely inseparable extensions of residue fields.

Proof. Part (1) follow from Lemma 0BRF with $I = JR$. If a is as in part (2), then a is the image of $x \in R^G \cap JR$. Hence $(T-x)^n = \prod_{\sigma \in G}(T - \sigma(x))$ is congruent to T^n modulo J by Lemma 0H34. This proves part (2). To see the final statement we may apply Algebra, Lemma 09EF. □

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